

Research Interests

- a. **Statistical Learning under Complex Data Conditions**
Semi-supervised learning, informative missingness, mixture models, robust estimation, kernel and regularisation methods
- b. **Computational Statistics and Optimization for Learning**
Efficient optimisation for statistical models, high-dimensional and non-convex problems, stochastic/iterative optimisation, evolutionary and swarm-based methods
- c. **Time-series, Forecasting, and Causal Inference**
Statistical modelling and uncertainty quantification in energy, environmental, and socio-economic systems; probabilistic forecasting; causal mechanisms in dynamical systems

Education

- 2022 **Doctor of Philosophy in Statistics**, *Queensland University of Technology*, Brisbane, Australia
2017 **Master of Applied Statistics**, *Lanzhou University*, Lanzhou, PR China
2014 **Bachelor of Economics in Economic Statistics**, *Anhui University*, Hefei, PR China

PhD Thesis

- Title *Statistical Support Vector Machines with Optimizations*
(The 2022 Executive Dean's Commendation for Outstanding Doctoral Thesis)
Supervisors Professors You-Gan Wang, Kevin Burrage, & Yu-Chu Tian

Employments

- Feb 2025– **Research Fellow**, *School of Mathematics and Physics, The University of Queensland*, St Lucia, Australia, with Professor Geoffrey J. McLachlan (Fellow of the Australian Academy of Science)
Dec 2022– Jan 2025 **Research Fellow**, *Faculty of Education and Arts, Australian Catholic University*, North Sydney, Australia, with Professor Herbert W. Marsh (Fellow of the Academy of the Social Sciences in Australia and the British Academy of Social Sciences)
Jul 2022– Nov 2022 **Associate Lecturer**, *School of Mathematical Sciences, Queensland University of Technology*, Brisbane, Australia, (short contract)
Jul 2021– Nov 2021 **Sessional Academic**, *School of Mathematical Sciences, Queensland University of Technology*, Brisbane, Australia
Jun 2018– Dec 2018 **Research Assistant**, *UQ Business School, The University of Queensland*, St. Lucia, Australia, with Dr Min Zhu

Refereed Articles (*Correspondence; Led study design & writing)

- [1] **Jinran Wu**, You-Gan Wang, and Geoffrey J McLachlan. Informative missingness and its implications in semi-supervised learning. *The Innovation Informatics*, 2:100033, 2026.
- [2] **Jinran Wu**, Xin Tian, You-Gan Wang, Tong Li, Qingyang Liu, Yayong Li, Lizhen Cui, Zhuangcai Tian, Jing Xu, Xianzhou Lyu, et al. AI ethics in geoscience: Toward trustworthy and responsible innovation. *Geography and Sustainability*, 7(1):100414, 2026.
- [3] **Jinran Wu**, Xin Tian, Qingyang Liu, Tong Li, Chanjuan Liu, Jing Xu, and Huida Zhao. Generative artificial intelligence and urban transport resilience: Rethinking sustainable mobility. *Safety Science*, 198:107148, 2026.

- [4] **Jinran Wu**, Herbert W. Marsh, Jiesi Guo, Johnmarshall Reeve, Reinhard Pekrun, Theresa Dicke, Hye-Ren Jang, and Geetanjali Basarkod. Mapping the intellectual landscape of educational psychology: Citation rankings and network structures of 60 journals, scholars, and institutions. *Learning and Individual Differences*, 125:102816, 2026.
- [5] Yongjun Xu, Zezhi Shao, Xin Liu, ..., **Jinran Wu**, et al. AI for science: Progress, challenges, and perspectives. *The Innovation*, 7, 2026.
- [6] Huanchao Zhou, Fang Wang, You-Gan Wang, **Jinran Wu**, Jiang Hu, Geoffrey J McLachlan, Zhi-dong Bai, and Jianghao Li. Signal plus noise type matrix in random matrix theory: A review. *Wiley Interdisciplinary Reviews: Computational Statistics*, 18(3):e70071, 2026.
- [7] Yayong Li, Zhengyi Du, Hong Zhang, Jonathan Wilton, **Jinran Wu**, Zongli Liu, and Nan Ye. Hierarchical cluster-based open-world graph active learning. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '26, page 935–945, 2026.
- [8] Yujia Huang, **Jinran Wu**, Zhe Ding, and Xi'an Li. Normalized Fourier induced coupled PINNs to solve the Dirichlet biharmonic equations in a large-scale domain. *Frontiers of Computer Science*, 20(7):2007354, 2026.
- [9] Xi'an Li, **Jinran Wu**, Xiao Ning, Yangshuai Wang, and Lei Zhang. Bayesian physics-informed neural networks to solve the partial differential equations with noise and incomplete constraints. *The Innovation Informatics*, 2:100039, 2026.
- [10] Tian Chen, Zhuangcai Tian, Xi'an Li, Jingxin Wu, Xiujun Guo, Yumeng Zhang, Haorui Wang, and **Jinran Wu**. Physics-informed neural networks help to strongly reduce uncertainty in ocean state forecasting. *The Innovation Informatics*, 2:100051, 2026.
- [11] Jing Xu, Siyuan Liu, **Jinran Wu**, Yuming Mo, Kun Wang, Qingfeng Ji, You-Gan Wang, and Ling Li. Multi-head attention driven aggregation-differentiation network for probabilistic groundwater depth forecasting and depth-stratified early warning: A multi-site efficient framework. *Environmental Modelling & Software*, 199:106934, 2026.
- [12] Chaojun Zou, Fang Wang, **Jinran Wu**, You-Gan Wang, and Xinghui Zhu. Rapid rapeseed identification with dual-graph discriminant analysis in hyperspectral imaging. *Chemometrics and Intelligent Laboratory Systems*, 277:105813, 2026.
- [13] Jichao Ma, Dandan Liu, Xi'an Li, Yayong Li, and **Jinran Wu***. A spatial-temporal normalization framework for fourier-enhanced physics-informed deep learning in large-scale wave propagation. *Discover Computing*, 2026.
- [14] Shuaiying Zhang, Zhen Dong, Huadong Lin, Zhendong Zhang, **Jinran Wu**, Sinong Quan, Wentao An, Tong Li, and Rajiv Pandey. Integrating linear and circular polarization features for polsar land cover classification with deep learning. *International Journal of Applied Earth Observation and Geoinformation*, 146:105090, 2026.
- [15] Yuchao Gao, **Jinran Wu**, Yang Yang, Zijin Wang, and Zhe Ding. Frequency-aware multi-task forecasting for integrated energy systems via variational mode decomposition and convolution-attention encoding. *IEEE Transactions on Smart Grid*, 17(1):818–831, 2026.
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- [17] Xin Tian, Qingyang Liu, Ram C. Dalal, Tong Li, **Jinran Wu***, Geoffrey J. McLachlan, Scott Chapman, and Yash P. Dang. Versioned soil organic carbon maps for climate-resilient and sustainable cities: A ridge-penalised linear mixed-effects ensemble of MIR-based predictions. *Physics and Chemistry of the Earth*, 143:104390, 2026.
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- [19] Zhuangcai Tian, Jiahao Tian, Sen Hua, Liyuan Yu, Hongwen Jing, and **Jinran Wu**. Microseismic source localization and application based on the attention-based deep feedforward neural network model. *Engineering Applications of Artificial Intelligence*, 177:114982, 2026.
- [20] Qingyang Liu, Xin Tian, Ram C. Dalal, **Jinran Wu**, Anquan Xia, Xiaoxuan Wu, Tong Li, Geoffrey J. McLachlan, Scott Chapman, and Yash P. Dang. Mapping soil organic carbon research in conservation agriculture: A systematic review. *Farming System*, 4(4):100241, 2026.
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- [22] Zhi Zheng, Siyuan Liu, Haohui Han, Shang Gao, Mumta Chhetri, **Jinran Wu**, Jun Ma, You-Gan Wang, Liya Fu, Weide Li, Haiyan Hu, and Chunji Liu. Grain yield loss caused by fusarium crown rot was mainly due to reduced numbers of fertile tillers in barley. *Farming System*, page 100256, 2026.
- [23] Zhuangcai Tian, Xiujun Guo, Jiaming Xiang, Tong Li, and **Jinran Wu**. Anthropogenic sustained increase in near bottom suspended particulate matter concentration of the deep marginal sea. *Environmental Research*, 293:123772, 2026.
- [24] Zihan Liu, Xiang Sui, Xianzhou Lyu, Xiaobiao An, **Jinran Wu**, Yuchen Lu, and Weijie Zhang. Multi-indicator water-quality prediction in mining areas using a feature-tokenizer transformer with spatiotemporal features. *Environmental Research*, 306:125210, 2026.
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- [26] Tong Li, Anquan Xia, Ram Dalal, Sean Manning, Deeksha Beniwal, Zhihong Xu, Rajiv Pandey, Lizhen Cui, Songchao Chen, Peng Fu, Yushu Xia, **Jinran Wu**, Yanfen Wang, and Yash Pal Dang. Mid-infrared spectroscopy for soil organic carbon estimation. Part I: Global review and meta-analysis. *Soil & Tillage Research*, 262:107236, 2026.
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- [32] Jichao Ma, Jiebin Luo, Dandan Liu, **Jinran Wu***, Yanjiang Wang, and Xi’an Li. Accurately modelling the resting-state functional connectivity with eigen-based connected-graph diffusion model. *Engineering Applications of Artificial Intelligence*, 178:115091, 2026.
- [33] Yang Yang, Zhihao Chen, Yuchao Gao, Zijin Wang, Zhe Ding, and **Jinran Wu**. Network traffic forecasting with transfer learning-based algorithm for long continuous missing data. *Expert Systems with Applications*, 297:129484, 2026.

- [34] Xing Zhang, **Jinran Wu**, Jingwen Wang, Tong Li, Yuchuan Fan, Weiguo Liu, and Jie Gao. Forest origin reorganizes environmental control of soil microbial diversity across natural and planted forests. *Ecological Indicators*, 189:115086, 2026.
- [35] Yujuan Zheng, Chengguang Zhong, Tong Lyu, **Jinran Wu**, Le Zhang, Jingwen Wang, Jie Gao, Tong Li, Yuchuan Fan, and Rajiv Pandey. Elevational patterns and thresholds of plant functional traits across China's biomes. *Ecological Indicators*, 185:114818, 2026.
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- [37] Zhuangcai Tian, Shengjie Rui, **Jinran Wu**, Ning Fan, and Lei Guo. Sustainability and environmental considerations in mining: From deep-sea to solid earth. *Frontiers in Earth Science*, 13:1712087, 2025.
- [38] Fengjing Cai, Yue Wang, Wanghua Yu, **Jinran Wu***, Chanjuan Liu, and Xi'an Li. ASISTGCRN: A novel approach to traffic prediction using attention-based spatiotemporal graph networks. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, page 09544070251390950, 2025.
- [39] Yang Yang, Yuchao Gao, **Jinran Wu***, and Shangce Gao. Optimal predictive load frequency control with multi-objective PID-based search algorithm. *Swarm and Evolutionary Computation*, 99:102214, 2025.
- [40] Zhesen Cui, Qingbin Zhang, Zhixing Zhao, Maosheng Yang, Pan Kong, Shuobao Cai, Jinran Zhang, Jing Xu, **Jinran Wu***, Rajib Pandey, and Tong Li. A two-stage probabilistic forecasting framework for dam displacement: Evidence from a Chinese watershed. *Journal of Hydrology: Regional Studies*, 62:102822, 2025.
- [41] **Jinran Wu***, Yang Yang, Shaolong Sun, and Yang Yu. Data-driven approaches for efficient smart grid systems. *Frontiers in Energy Research*, 12:1536459, 2025.
- [42] Yayong Li, Xubo Zhang, Hong Zhang, Nan Ye, Zongli Liu, and **Jinran Wu**. Generalized few-shot node classification via training set refinement. In *Pacific Rim International Conference on Artificial Intelligence*, pages 492–508. Springer, 2025.
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- [44] Jing Xu, Yuming Mo, Qihao Jiang, Lingzhong Kong, **Jinran Wu**, Zhe Ding, Guangqiu Jin, and Ling Li. Rainfall-dependent influence of water parameter interactions and land use on lake water quality: A hybrid ensemble approach and management implications. *Journal of Hydrology*, 662, Part B:134019, 2025.
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- [46] Fengjing Cai, Yue Wang, Zhuangcai Tian, Xi'an Li, Jing Xu, Yuming Mo, Shaotong Zhang, and **Jinran Wu**. Integrating attention mechanisms in graph neural networks for marine oil spill detection. *Journal of Ocean University of China*, 24:1327—1340, 2025.
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- [54] Yu-Qi Yin, Chanjuan Liu, and **Jinran Wu**. Scheduling automated guided vehicles considering the buffer lane constraints. *Engineering Computations*, 42(4):1641–1670, 2025.
- [55] Shangrui Zhao, Weiqi Yu, **Jinran Wu***, Xi'an Li, and You-Gan Wang. An adaptive regression algorithm with a clustering process for multi-modal data prediction. *Discover Computing*, 28(1):94, 2025.
- [56] Xi'an Li, **Jinran Wu**, Yujia Huang, Zhe Ding, Xin Tai, Liang Liu, and You-Gan Wang. Fourier-feature induced physics informed randomized neural network method to solve the biharmonic equation. *Journal of Computational and Applied Mathematics*, 468:116635, 2025.
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- [58] Yang Yang, Yuchao Gao, Shangce Gao, and **Jinran Wu***. Optimizing PID controllers for multi-area automatic generation control with improved NSGA-II. *CAAI Transactions on Intelligence Technology*, 10:1135—1147, 2025.
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Review Services

Grants

- DAAD Postdoc-Net-AI under German Academic Exchange Service

Journals

- | | |
|---|--|
| ○ Agronomy | ○ Ain Shams Engineering Journal |
| ○ Algorithms | ○ Applied Artificial Intelligence |
| ○ Applied Intelligence | ○ Applied Mathematics and Mechanics |
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| ○ Asia Pacific Journal of Marketing and Logistics | ○ Automatika |
| ○ Biology and Fertility of Soils | ○ CAAI Transactions on Intelligence Technology |
| ○ Canadian Journal of Chemical Engineering | ○ Carbon Footprints |
| ○ Clinical Cardiology | ○ Communications in Computational Physics |
| ○ Complex & Intelligent Systems | ○ Concurrency and Computation: Practice and Experience |
| ○ Control Engineering Practice | ○ Cybernetics and Systems |
| ○ Data Science and Management | ○ Earth's Future |
| ○ Earthquake Engineering and Resilience | ○ Electric Power Systems Research |
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- Engineering Applications of Artificial Intelligence
- Engineering Computations
- Engineering Research Express
- Environmental Engineering Science
- Environmental Modeling and Assessment
- Environmental Science: Advances
- Expert Systems with Applications
- Forecasting
- Frontiers of Computer Science
- Frontiers of Structural and Civil Engineering
- Human and Ecological Risk Assessment
- IEEE Antennas and Wireless Propagation Letters
- IEEE Geoscience and Remote Sensing Letters
- IEEE Internet of Things Journal
- IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Cybernetics
- IEEE Transactions on Emerging Topics in Computational Intelligence
- IEEE Transactions on Geoscience and Remote Sensing
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Knowledge and Data Engineering
- IEEE Transactions on Network and Service Management
- IEEE Transactions on Pattern Analysis and Machine Intelligence
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- IEEE/ACM Transactions on Audio, Speech, and Language Processing
- IET Communications
- IET Generation, Transmission & Distribution
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- Information Processing in Agriculture
- Integrated Environmental Assessment and Management
- International Journal for Numerical Methods in Engineering
- International Journal of Ambient Energy
- International Journal of Computer Mathematics
- International Journal of Forecasting
- Engineering Applications of Computational Fluid Mechanics
- Engineering Optimization
- Entropy
- Environmental Impact Assessment Review
- Environmental Science and Pollution Research
- Europhysics Letters
- F1000Research
- Frontiers in Energy Research
- Frontiers of Information Technology & Electronic Engineering
- Geo-Spatial Information Science
- IEEE Access
- IEEE Computational Intelligence Magazine
- IEEE IAS Publications
- IEEE Journal of Biomedical and Health Informatics
- IEEE Sensors Journal
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Dependable and Secure Computing
- IEEE Transactions on Fuzzy Systems
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Intelligent Vehicles
- IEEE Transactions on Microwave Theory and Techniques
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Reliability
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE Wireless Communications Letters
- IEEE/CAA Journal of Automatica Sinica
- IET Electric Power Applications
- IET Power Electronics
- IET Smart Grid
- Innovation Informatics
- International Journal for Numerical and Analytical Methods in Geomechanics
- International Journal of Adaptive Control and Signal Processing
- International Journal of Computational Intelligence Systems
- International Journal of Environmental Science and Technology
- International Journal of Green Energy

- International Journal of Information Technology & Decision Making
- International Journal of Numerical Methods for Heat & Fluid Flow
- International Journal of Urban Sciences
- JMIR Public Health and Surveillance
- Journal of Applied Statistics
- Journal of Chemical & Engineering Data
- Journal of Computer Science and Technology
- Journal of Ecohydraulics
- Journal of Field Robotics
- Journal of Groundwater Science and Engineering
- Journal of Marine Engineering and Technology
- Journal of Modern Power Systems and Clean Energy
- Journal of Vibration and Control
- Machine Learning: Science and Technology
- Marine Georesources and Geotechnology
- Measurement Science and Technology
- Medicine
- Network: Computation in Neural Systems
- Neurocomputing
- Nonlinear Dynamics
- Ocean Engineering
- Personnel Review
- Physica Scripta
- Physics of Fluids
- Results in Chemistry
- Science Progress
- Ships and Offshore Structures
- SN Computer Science
- Spectroscopy Letters
- Statistics in Biosciences
- Symmetry
- Technological Forecasting and Social Change
- Transactions on Emerging Telecommunications Technologies
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- International Journal of Systems Science
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- Journal of Ambient Intelligence and Humanized Computing
- Journal of Bionics Engineering
- Journal of Computational Social Science
- Journal of Earth Science
- Journal of Environmental Management
- Journal of Geophysical Research: Oceans
- Journal of Hospitality Marketing & Management
- Journal of Mechanical Engineering Science
- Journal of Supercomputing
- Machine Learning
- Malaysian Journal of Mathematical Sciences
- Mathematics
- Mechanics of Advanced Materials and Structures
- Microchemical Journal
- Neural Computing & Applications
- New England Journal of Medicine
- Numerical Methods for Partial Differential Equations
- PeerJ Computer Science
- Petroleum Science and Technology
- Physical Chemistry Chemical Physics
- PLOS ONE
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- [12] Qiu, X., Yuan, J., **Wu, J.**, Li, W., Guo, J., & Leung, F. (2026). Leveraging Machine Learning for Clustered Data: Model Development and Application to Examining the Impact of COVID-19 on Student Mathematics Achievement. *International Meeting of the Psychometric Society 2026*. Seoul, Republic of Korea
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- [9] Zhang, S., **Wu, J.**, & Zhao, Z. (2023). Estimating sediment transport parameters from field-measured suspended fine concentration time series with a new conceptual model. *2023 The XIV Congress of the International Association for Engineering Geology and the Environment*. Chengdu, China
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- [6] Cui, Z., **Wu, J.**, & Lian, W. (2021). A modified gated recurrent unit neural network for crude oil price forecasting. *The 2021 International Academic Conference on Machine Learning, Big Data, and Statistics, Financial Information Technology (ICMBSF 2021)*. Brisbane, Australia
- [5] Gao, Y., Qian, C., Tao, Z., Zhou, H., **Wu, J.**, & Yang, Y. (2020). Improved whale optimisation algorithm via cellular automata. *The 2020 International Conference on Progress in Informatics and Computing (PIC-2020)*. Shanghai, China
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- [2] **Wu, J.**, & Ding, Z. (2020). Improved grey model by dragonfly algorithm for Chinese tourism-demand forecasting. *The 33th International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems*. Kitakyushu, Japan
- [1] **Wu, J.**, & Wang, Y.-G. (2019). Time series regression modelling using phase space reconstruction and deep neural networks. *The 20th INFORMS Applied Probability Society Conference*. Brisbane, Australia

Professional Memberships and Academic Services

Jul 2024–	Academic Editor of the <i>PloS One</i>
May 2026–	Associate Editor of <i>Journal of Electrical Systems and Information Technology</i>
Jan 2026–	Editorial Board Member of the <i>Applied Mathematics and Statistics</i>
Oct 2025–	Youth Editor of <i>The Innovation Informatics</i>
Dec 2023–	Youth Editor of <i>Deep Underground Science & Engineering</i>
Jun 2026	Guest Editor, the Special Issue on “Particulate and Granular Flow in the Ocean” of <i>Physics of Fluids</i>
Jan 2026	Guest Editor, the Special Issue on “Artificial Intelligence for Biochar Applications in Agricultural, Forest, and Grassland Ecosystems: From Soil Carbon to Ecosystem Sustainability” of <i>Biochar</i>
Sep 2025	Guest Editor, the Special Issue on “Artificial Intelligence in Physical Geography: Theory, Methods, and Frontiers of Practice” of <i>Geographical Science</i> [In Chinese]
Jun 2025	Guest Editor, the Special Issue on “Foundation Models and Artificial Intelligence in Geography and Sustainability” of <i>Geography and Sustainability</i>
May 2025	Guest Editor, the Special Issue on “Impact of Generative Artificial Intelligence on Transportation Safety and Resilience” of <i>Safety Science</i>
Apr 2025	Guest Editor, the Special Issue on “Stability and Persistence of Soil Organic Carbon Pool in a Changing World” of <i>Biology and Fertility of Soils</i>
Mar 2025	Member of the Program Committee of 26th International Conference on Artificial Intelligence in Education (AIED 2025), Palermo, Italy
Nov 2024	Member of the Program Committee of the Australasian Data Science and Machine Learning Conference (AusDM'24), Melbourne, Australia

- Jul 2024 Member of the Program Committee of the 17th International Conference on Educational Data Mining (EDM 2024), Atlanta, USA
- Sep 2024 Guest Editor, the Special Issue on “Deep-Sea Sediment Transport and Mining” of *Frontiers in Earth Science*
- Jan 2024 Guest Editor, the Special Issue on “Data-Driven Approaches for Efficient Smart Grid Systems” of *Frontiers in Energy Research*
- Sep 2023 Session Convener, the XIV Congress of the International Association for Engineering Geology and the Environment, Chengdu, China
- Jan 2023 Guest Editor, the Special Issue on “Machine Learning in Environmental Modelling” of *Environmental Modeling and Assessment*
- Jul 2022– Review Editor, Editorial Board of Machine Learning and Artificial Intelligence (speciality section of *Frontiers in Big Data* and *Frontiers in Artificial Intelligence*)

Teaching Experience

- 2026se1 **Instructor** for MATH7014 (Minor Project), UQ
- 2025se2 **Instructor** for MATH7012 (Project or Thesis), UQ
- 2025se2 **Guest Lecturer** for STAT4401 (Advanced Statistics), UQ
- 2024se2 **Coordinator** for EDCP600 (Interpreting Literature and Data), ACU
- 2022se2 **Coordinator** for MXB343 (Modelling Dependent Data), QUT
- 2022se1 **Guest Lecturer** for MXN441 (Advanced Statistical Inference and Modelling), QUT
- 2021se2 **Tutor** for MXB343 (Modelling Dependent Data), QUT
- 2021se1 **Guest Lecturer** for MXN441 (Advanced Statistical Inference and Modelling), QUT

Grants and Awards

- 2024 The 2024 Early Career Researcher Grant, Faculty of Education & Arts, ACU
Be awarded the amount of AUD\$5,940.24
- 2023 The 2022 Executive Dean's Commendation for Outstanding Doctoral Thesis, Faculty of Science, QUT
- 2022 The Chinese Government Award for Outstanding Self-financed Students Abroad
The highest governmental award to Chinese graduate students studying abroad
- 2020 The QUT Science and Engineering Faculty's High Achiever Award
- 2018 The Australian Government Research Training Program Scholarship (International)
- 2018 The ACEMS Top Up Scholarship